

Nonequilibrium population dynamics under evolving resource inequality

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Main equations (see details in manuscript)

$$\frac{\partial N}{\partial t} = \left(b \cdot I_{\frac{1}{2}} \left(\frac{k(t) \cdot \varepsilon_b \cdot N}{R(N)} + 1, k(t) \right) + (1 - \delta) \cdot I_{\frac{1}{2}} \left(\frac{k(t) \cdot \varepsilon_m \cdot N}{R(N)} + 1, k(t) \right) - 1 \right) \cdot N$$

$$\frac{\partial k}{\partial t} = \mu(k)$$

1.1 Condorcet model ($R(N) = \varepsilon \cdot N$, $\mu(k) = \mu$)

$$\frac{\partial N}{\partial t} = \left(b \cdot I_{\frac{1}{2}} \left(\frac{k(t) \cdot \varepsilon_b}{\varepsilon_c} + 1, k(t) \right) + (1 - \delta) \cdot I_{\frac{1}{2}} \left(\frac{k(t) \cdot \varepsilon_m}{\varepsilon_c} + 1, k(t) \right) - 1 \right) \cdot N$$

$$\frac{\partial k}{\partial t} = \mu$$

Parameters

$\sigma := \frac{1}{90}$	Aging mortality rate (inverse of potential lifespan)
$\varepsilon_{aa} := 0.0$	Additional mortality rate of birth givers
$b := 0.23$	Fecundity rate, number of offspring per reproductive individual per time, for sexual population $\sim 0.5 \cdot (\text{potential fertility} / \text{reproductive lifespan})$
$\varepsilon_b := 1.2$	Resource rate necessary for reproduction
$\varepsilon_m := 0.5$	Resource rate necessary for survival
$\varepsilon_c := 1.0$	Resource productivity, incoming resource rate per individual
$\mu := 0.2$	Equality change rate
$N_0 := e^{0.5}$	Initial population size

1.2 Condorcet model (calculations)

The free variable **k** is the level of equality in resource distribution.

$$B(k, ec) := (b - c) \cdot \text{ibeta}\left(\frac{1}{2}, eb \cdot \frac{k}{ec} + 1, k\right)$$

Birth rate

$$M(k, ec) := 1 - (1 - \sigma) \cdot \text{ibeta}\left(\frac{1}{2}, em \cdot \frac{k}{ec} + 1, k\right)$$

Mortality rate

$$\text{LRS}(k, ec) := \frac{B(k, ec)}{M(k, ec)}$$

LRS (Lifetime Reproductive Success)

$$\text{G}(k, ec) := B(k, ec) - M(k, ec)$$

Growth rate

$$\text{kb}(z) := \begin{cases} k \leftarrow \text{root}(\text{G}(z, ec), z) \\ \text{return } k \end{cases}$$

The range of equality [kl,kr] where the growth rate is positive

$$kl := \text{kb}(1) \quad kr := \text{kb}(200)$$

$$\text{Mkb}(z) := \begin{cases} k \leftarrow \text{root}\left(\frac{d}{dz} \text{G}(z, ec), z\right) \\ \text{return } k \end{cases}$$

The value of the equality km at which the growth rate is maximal

$$\text{km} := \text{Mkb}(kl)$$

$$\text{Mlrs}(z) := \begin{cases} k \leftarrow \text{root}\left(\frac{d}{dz} \text{LRS}(z, ec), z\right) \\ \text{return } k \end{cases}$$

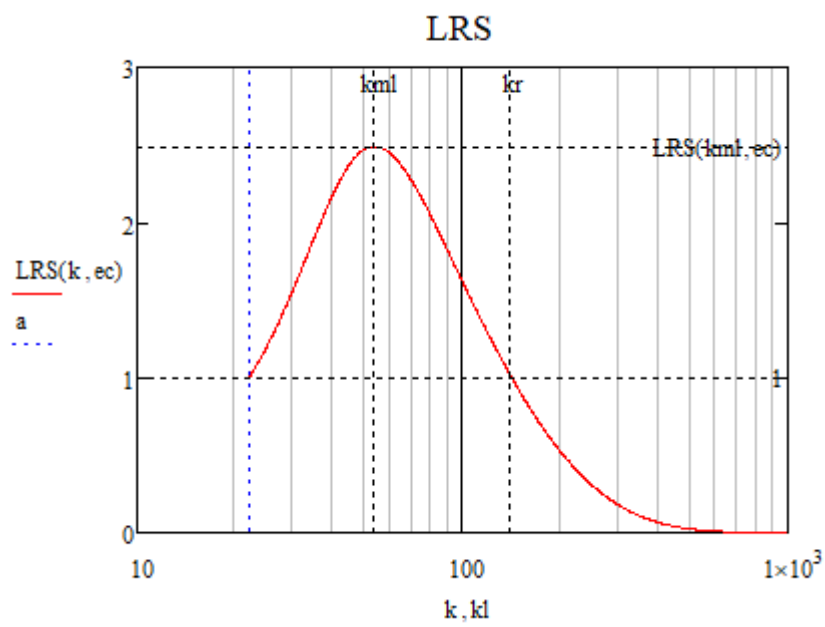
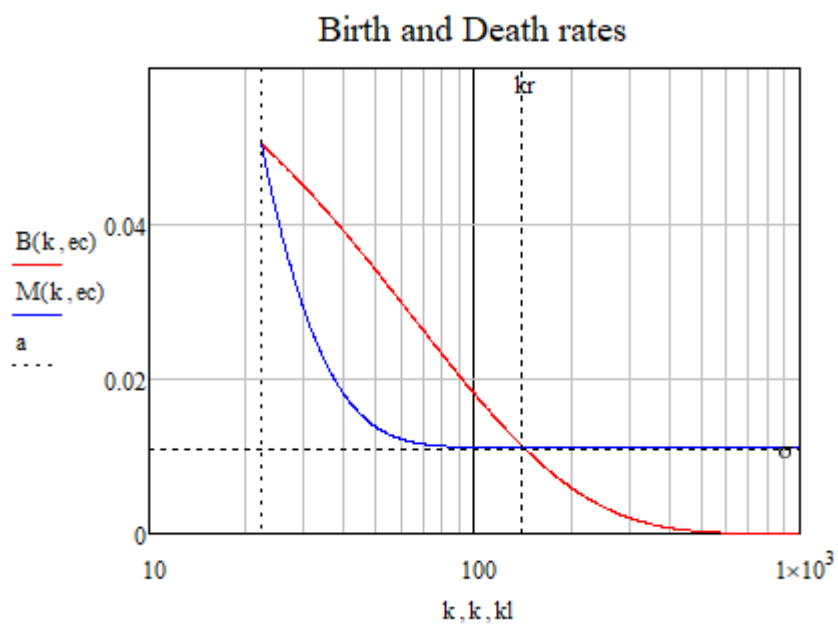
The value of the equality kml at which LRS is maximal

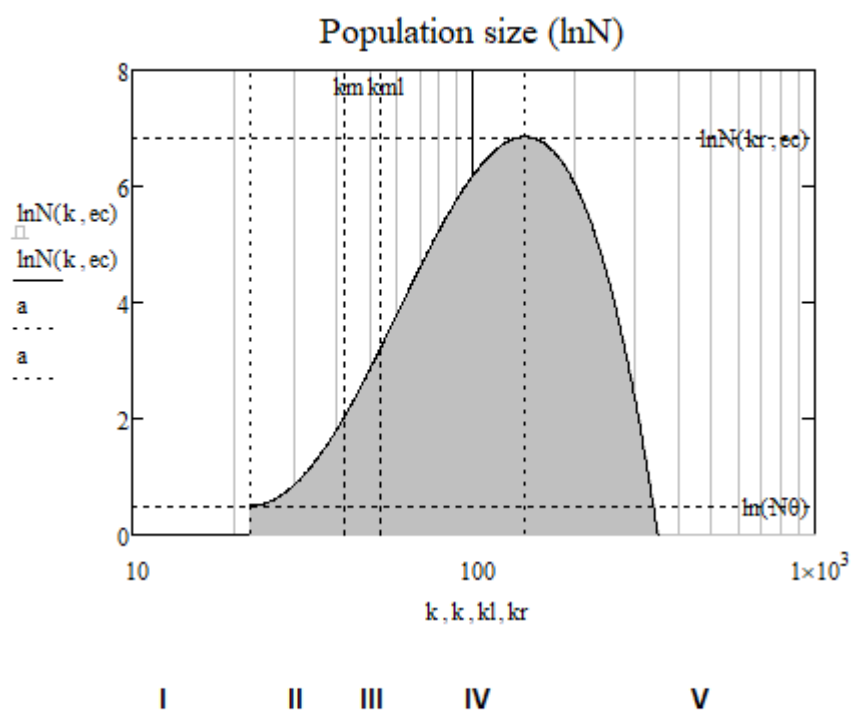
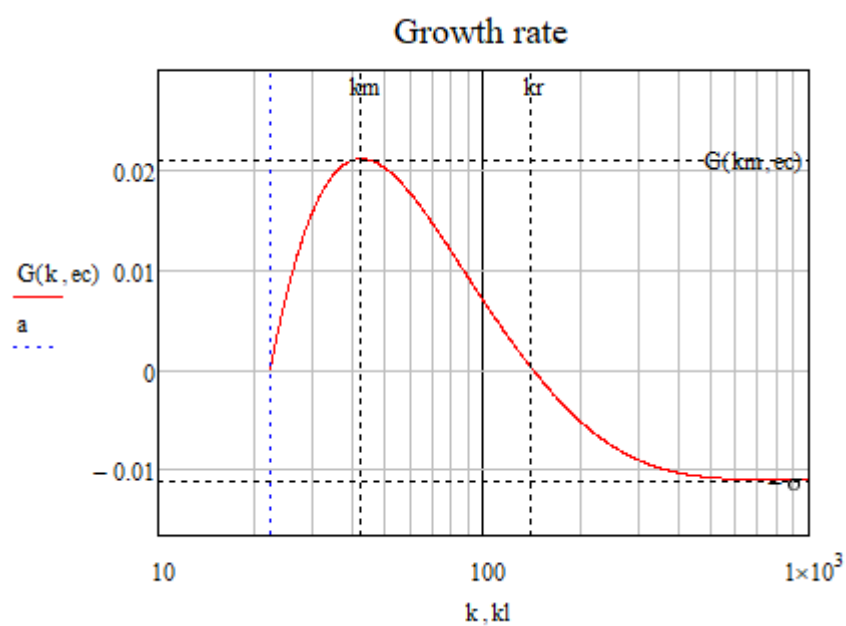
$$\text{kml} := \text{Mlrs}(2 \cdot kl)$$

$$\ln N(k, ec) := \begin{cases} \ln(N_0) & \text{if } k < kl \\ \left(\ln(N_0) + \int_{kl}^k \frac{\text{G}(z, ec)}{\mu} dz \right) & \text{otherwise} \end{cases}$$

Log population size

1.3 Condorcet model (plots)



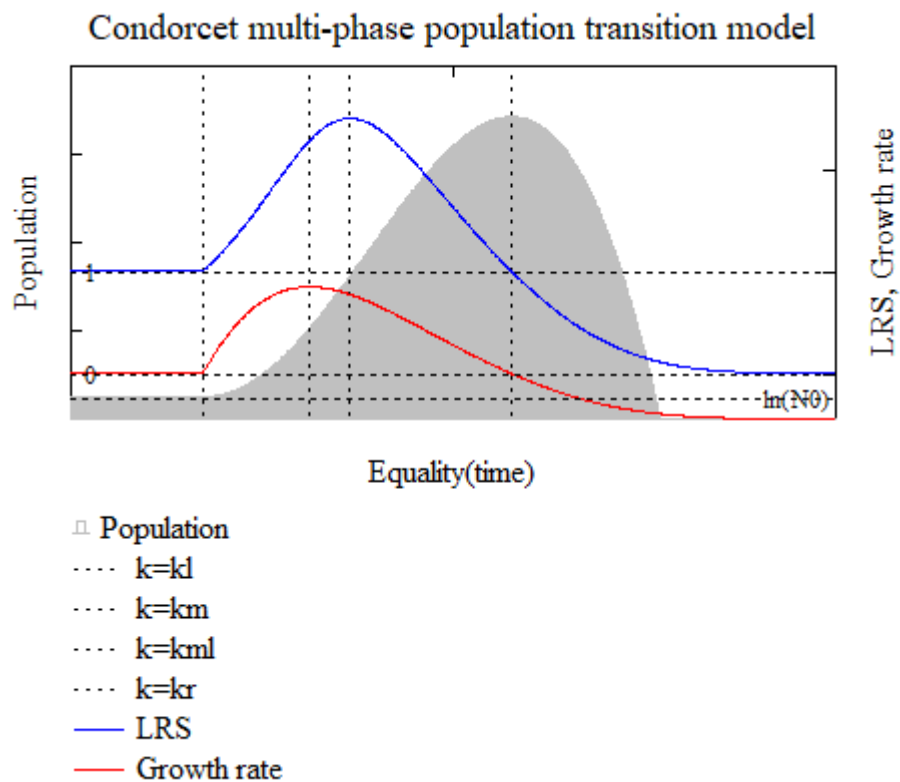


1.4 Condorcet multi-phase population transition model

Phases	Range of equality	Population growth
I (Malthsian)	$]0, k_l]$	close to zero
II (Acceleration)	$[k_l, k_m]$	expotential with increasing rate
III (Exponential)	$[k_m, k_{ml}]$	expotential with constant rate
IV (Retardation)	$[k_{ml}, k_r]$	expotential with deceasing rate
V (Decline)	$[k_r, \text{inf}[$	expotential with negative rate

$$\text{LRSI}(k, \text{ec}) := \begin{cases} 1 & \text{if } k < k_l \\ \text{LRS}(k, \text{ec}) & \text{otherwise} \end{cases}$$

$$\text{GI}(k, \text{ec}) := \begin{cases} 0 & \text{if } k < k_l \\ \text{G}(k, \text{ec}) & \text{otherwise} \end{cases}$$



2.1 Transient Malthusian model ($R(N)=R$, $\mu(k)=\mu$)

$$\frac{\partial N}{\partial t} = \left(b \cdot I_{\frac{1}{2}} \left(\frac{k(t) \cdot \varepsilon_b \cdot N}{R} + 1, k(t) \right) + (1 - \sigma) \cdot I_{\frac{1}{2}} \left(\frac{k(t) \cdot \varepsilon_m \cdot N}{R} + 1, k(t) \right) - 1 \right) \cdot N$$

$$\frac{\partial k}{\partial t} = \mu$$

Parameters

$R := 100$ Fixed resource amount

$b := 1.0$ The remaining parameters are the same as specified above

The free variables k and n represent the level of equality and the population size.

2.2 Transient Malthusian model (calculations)

$$B(n, k) := (b - c) \cdot \text{ibeta} \left(\frac{1}{2}, \frac{eb \cdot k \cdot n}{R} + 1, k \right) \quad \text{Birth rate}$$

$$M(n, k) := 1 - (1 - \sigma) \cdot \text{ibeta} \left(\frac{1}{2}, \frac{em \cdot k \cdot n}{R} + 1, k \right) \quad \text{Morality rate}$$

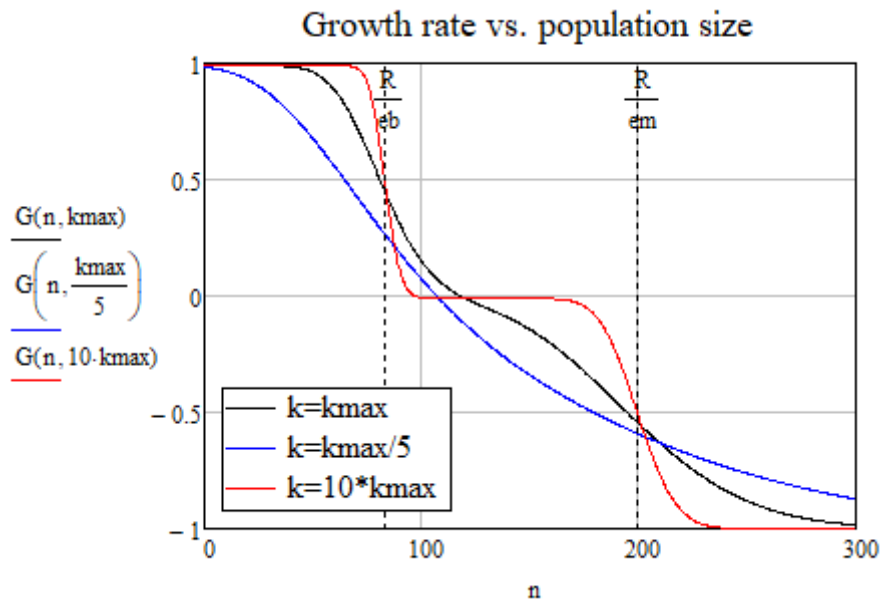
$$LRS(n, k) := \frac{B(n, k)}{M(n, k)} \quad \text{LRS = 1 at the equilibrium population size } N^*$$

$$G(n, k) := B(n, k) - M(n, k) \quad \text{Growth rate = 0 at the equilibrium population size } N^*$$

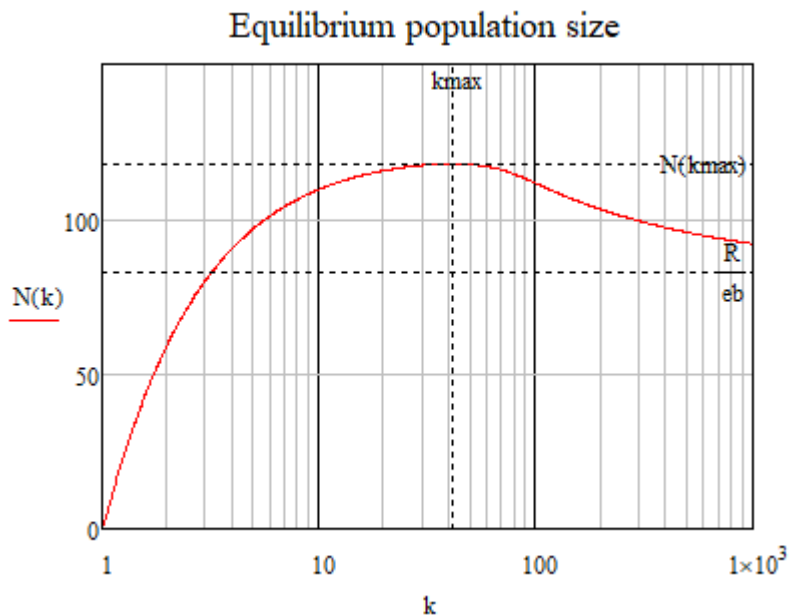
$$N(k) := \text{root} \left(G(n, k), n, eb, \frac{R}{em} \right) \quad \text{Equilibrium population size } N^*(k)$$

$$k_{\max} := \text{root} \left(\frac{d}{dk} N(k), k, eb, \frac{R}{em} \right) \quad \text{The level of equality at which the population size reaches its maximum}$$

2.3 Transient Malthusian model (plots)



Note: As equality increases, the population becomes increasingly density-independent within the interval $[R/eb, R/em]$. This means it can have any population size within this interval with nearly equal probability. In this case, it is more appropriate to use the term "equilibrium population interval" rather than "equilibrium point".



Note: In equilibrium-based models (i.e., where $B(n, k) \approx M(n, k)$ always holds), there is an optimal level of equality $k_{\max}$ at which the population reaches its maximum size.

3.1 Malthusian model ($R(N)=\epsilon \cdot r^* \ln(N)$, $\mu(k)=0$)

$$\frac{\partial N}{\partial t} = \left(b \cdot I_{\frac{1}{2}} \left(\frac{k \cdot \epsilon_b \cdot N}{\epsilon_r \cdot \ln(N)} + 1, k \right) + (1 - \delta) \cdot I_{\frac{1}{2}} \left(\frac{k \cdot \epsilon_m \cdot N}{\epsilon_r \cdot \ln(N)} + 1, k \right) - 1 \right) \cdot N$$

$$\frac{\partial k}{\partial t} = 0$$

Parameters

$\epsilon_r := 20$ Resource growth rate coefficient

$\epsilon_b := 1.0$ The remaining parameters are the same as specified above

The free variables **k** and **n** represent the level of equality and the population size.

3.2 Malthusian model (calculations)

$$B(n, k) := (b - c) \cdot \text{ibeta} \left(\frac{1}{2}, \frac{\epsilon_b \cdot k \cdot n}{\epsilon_r \cdot \ln(n)} + 1, k \right) \quad \text{Birth rate}$$

$$M(n, k) := 1 - (1 - \sigma) \cdot \text{ibeta} \left(\frac{1}{2}, \frac{\epsilon_m \cdot k \cdot n}{\epsilon_r \cdot \ln(n)} + 1, k \right) \quad \text{Morality rate}$$

$$LRS(n, k) := \frac{B(n, k)}{M(n, k)} \quad \text{LRS = 1 at the equilibrium population size } N^*$$

$$G(n, k) := B(n, k) - M(n, k) \quad \text{Growth rate=0 at the equilibrium population size } N^*$$

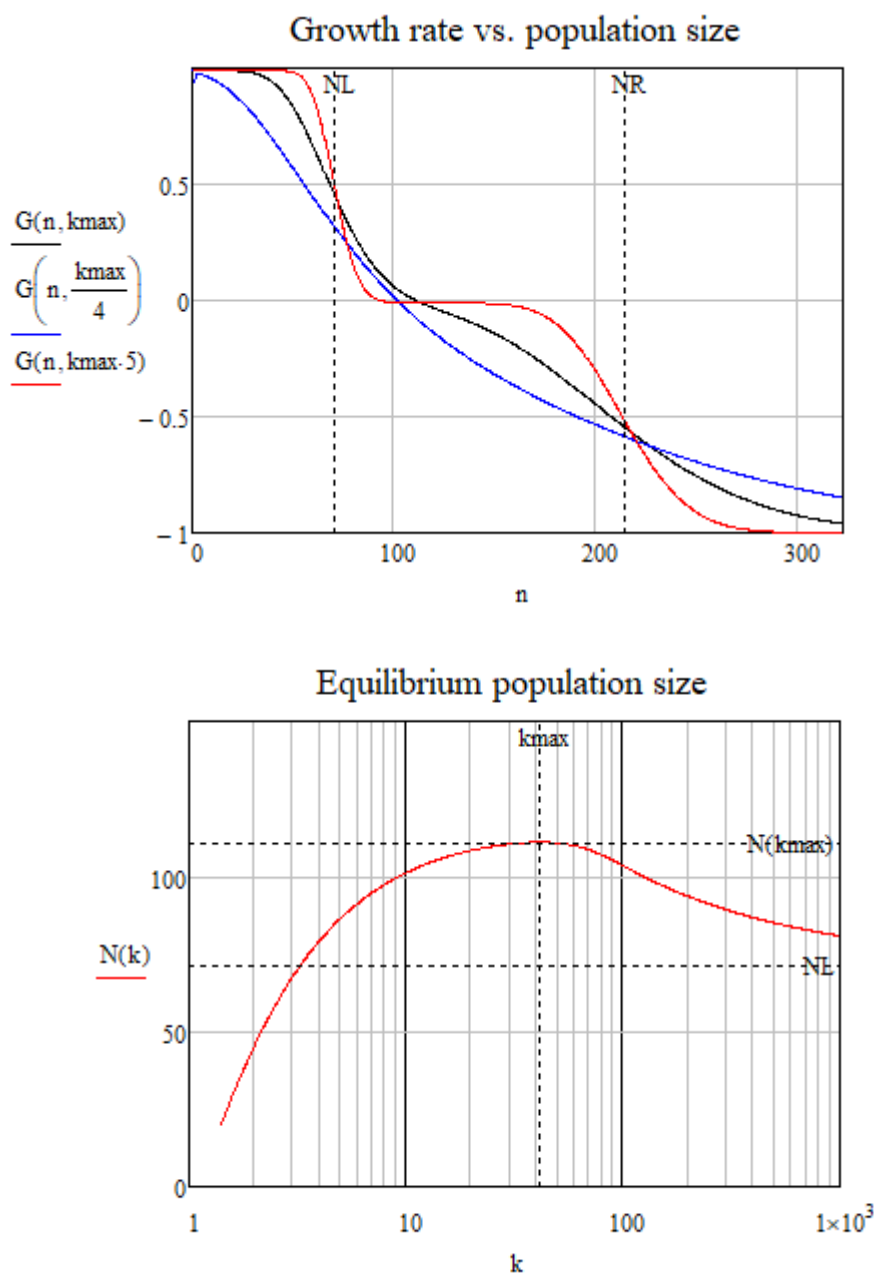
$$Nb(x) := \begin{cases} y \leftarrow \epsilon_r \\ \text{return } \text{root} \left(\frac{\ln(1 + y)}{y} - \frac{x}{\epsilon_r}, y \right) \end{cases} \quad \text{Asymptotic population size intervals at which the growth rate is maximal [0, NL] and at which the growth rate is close to zero [NL, NR] (i.e. equilibrium population interval).}$$

$$NL := Nb(\epsilon_b) \quad NR := Nb(\epsilon_m)$$

$$N(k) := \text{root}(G(n, k), n, \epsilon_b, NR) \quad \text{Equilibrium population size } N^*(k)$$

$$z := NL \quad k_{\max} := \text{root} \left(\frac{d}{dz} N(z), z \right) \quad \text{The level of equality at which the population size reaches its maximum}$$

3.3 Malthusian model (plots)



Note: An equilibrium arises given any sublinear relationship between the resource and the population size. Under these conditions, the results remain qualitatively the same.